

SECURECHAT

AI-Powered Zero-Trust Secure Messaging Platform

Document Type: Software Requirements Specification + Product/System Specification

Version: 1.0

Status: Proposed / Architecture Baseline

Platform: Android-first, Web/Desktop, future iOS

Core Philosophy: Privacy by Design + Zero Trust + AI-Assisted Security

1. Executive Summary

SecureChat is a privacy-focused instant messaging application designed to provide the familiar communication experience of modern messaging platforms while adding an intelligent, personalized security agent to every user's communication environment.

Unlike conventional messaging applications, SecureChat treats every incoming message, link, attachment, contact, and conversational interaction as untrusted by default.

The system identifies phishing attempts, malicious URLs, scams, social engineering, psychological manipulation, credential harvesting, impersonation, financial fraud, suspicious attachments, requests for sensitive information, potential data leakage, and account takeover attempts.

The system provides an intuitive security indicator on messages:

- Green: No significant security concern detected
- Orange: Suspicious / uncertain / requires caution
- Red: High-confidence malicious or dangerous content

2. Product Vision

SecureChat's vision is:

"Make secure communication as effortless as ordinary messaging."

Users should not need to understand phishing, social engineering, malicious URLs, manipulation techniques, OPSEC, data leakage, or identity spoofing. The security agent works silently in the background and intervenes only when necessary.

3. Core Design Principle

Zero Trust Communication Model

Never automatically trust a message, sender, link, attachment, or conversation merely because it appears familiar.

Identity → Context → Message → URL / Attachment → Behavior → Risk Analysis → User Decision

A known contact can still have a compromised account, send a malicious link, unknowingly forward malware, be impersonated, be socially engineered, or accidentally expose confidential information.

4. Major Product Components

SECURECHAT consists of:

- Messaging
- Security AI
- WhatsApp Bridge / SecureBridge
- Privacy Engine
- Zero-Trust Engine
- Security Center

Messaging includes chats/DMs, groups, calls (later phase), and media.

Security AI includes threat engine, context engine, phishing engine, manipulation AI, and privacy/DLP analysis.

SecureBridge analyzes supported third-party notification content.

5. User Types

5.1 Normal User

Can create an account, communicate, create groups, send media, receive security warnings, configure AI, control privacy, exclude selected chats, manage trusted contacts, and view security explanations.

5.2 Security-Conscious User

Additional capabilities include advanced threat analysis, detailed risk explanations, security logs, threat history, URL inspection, sender reputation, manipulation analysis, and privacy leak detection.

5.3 Administrator

Manages infrastructure, abuse reports, system health, threat intelligence, model versions, moderation policies, and security policies. Administrators should not automatically have access to users' private message contents.

6. Registration & Authentication

SecureChat shall support:

- Phone registration
- Email registration
- Username
- Optional biometric authentication
- Device registration
- Device verification
- Multi-device sessions
- Account recovery

Recommended flow:

Phone/Email → Identity Verification → Device Registration → Cryptographic Identity → Secure Session

7. End-to-End Encryption

SecureChat should use strong end-to-end encryption for messages.

User A → Encrypt locally → Encrypted message → Server → Encrypted message → User B → Decrypt locally

Security requirements:

- Forward secrecy
- Secure key exchange
- Device keys
- Session keys
- Key rotation
- Device revocation
- Identity verification
- Encrypted local database
- Encrypted backups

Use well-reviewed cryptographic protocols and primitives; do not invent a custom encryption protocol.

8. Personal AI Security Agent

Every user gets a personal security agent, SecureGuard AI.

The agent maintains a security context per conversation:

- Chat: Family → Security Context A
- Chat: University → Security Context B
- DM: Contact → Security Context C
- Group: Project → Security Context D

Contexts must remain logically isolated.

9. Conversation Context Engine

The AI should understand conversational context rather than analyzing isolated messages.

For example, a sequence involving an account problem, an urgent link, password entry, secrecy, and pressure should be analyzed as a combined pattern.

10. AI Security Pipeline

Incoming Message → Preprocessing → Message Classification → Context Retrieval → URL Analysis → Attachment Analysis → Sender Trust Analysis → Social Engineering Analysis → Sensitive Data Analysis → Risk Aggregation → Confidence Calculation → Security Decision → User Interface

11. Risk Classification

Use a risk score from 0 to 100.

Score	Classification
0–24	Green / Safe
25–59	Orange / Suspicious
60–79	Orange / High suspicion
80–100	Red / Critical

Thresholds should be calibrated through testing rather than treated as absolute truth.

12. Green Security Indicator

Every analyzed low-risk message can display a small green indicator in the lower-right portion of the message bubble.

Clicking it can show:

- Security Status: Low Risk
- No suspicious URL detected
- No sensitive-data request detected
- Sender is known
- No significant social-engineering pattern detected

13. Orange Warning

For suspicious messages, show an orange indicator and a warning explaining:

- Urgency detected
- Account-related claim
- External verification link
- Limited sender trust history

Actions:

- Inspect Link
- Continue Anyway
- Report
- Block Sender
- Ask AI

14. Red Critical Warning

For high-confidence malicious content, display a prominent warning.

Example:

HIGH-RISK MESSAGE

This message appears to contain a phishing attempt.

Risk: 94/100

Actions:

- Block link
- Report message
- Block sender
- Delete message
- View analysis
- Continue anyway (advanced/user-controlled where appropriate)

15. Phishing Detection Engine

Analyze:

- Suspicious domains
- Lookalike domains
- Unicode homographs
- Shortened URLs
- Excessive redirects
- Suspicious subdomains
- URL obfuscation
- Credential collection pages
- Newly registered domains where intelligence is available
- Mismatched display/domain information
- Suspicious path/query structures

16. URL Reputation

Combine:

Domain Reputation + Threat Intelligence + DNS/Domain Signals + Certificate Signals + URL Structure + Redirect Analysis + Page Behavior + AI Classification

The system should provide a threat confidence score.

17. "Guess or Really Check" Requirement

The system must distinguish between heuristic suspicion and verified evidence.

Detection basis can include:

- AI inference
- Threat intelligence
- URL inspection
- Domain reputation

- Sandbox analysis
- Conversation context

The system must never claim that something was verified if it only inferred it.

18. Social Engineering Detection

Analyze observable patterns associated with:

- Urgency
- Fear
- Authority impersonation
- Reward manipulation
- Guilt
- Intimidation
- Secrecy requests
- Emotional pressure
- Isolation
- Artificial scarcity
- Financial pressure
- Credential requests
- Authority claims
- Relationship exploitation
- Trust exploitation

19. Psychological Manipulation Engine

The system should identify observable conversational patterns, not diagnose people or label individuals as psychologically manipulative.

Good: "This conversation contains repeated urgency and secrecy pressure."

Bad: "This person is a manipulative person."

20. Conversation Security Summary

Every chat can have an AI-generated security description.

Example:

Current status: Green / Low Risk

Summary: A discussion about a university project. No suspicious requests detected.

Observations:

- No phishing links
- No credential requests
- No financial requests

- No major manipulation indicators
- Participant identities unchanged

21. Dynamic Conversation Description

The security description updates as the conversation changes:

Normal conversation → External link introduced → Urgency detected → Credential request → Phishing confirmed

22. Secret / Sensitive Information Protection

Detect:

- Passwords
- OTPs
- API keys
- Access tokens
- Credit/debit card information
- Identity numbers
- Private addresses
- Confidential documents
- Private keys
- Recovery codes
- Authentication codes
- Sensitive business information

Warn before sending potentially sensitive content.

23. Data Leakage Prevention

Recognize patterns such as:

- AWS access keys
- GitHub tokens
- JWTs
- Private keys
- Database credentials
- Passwords
- OTPs
- API tokens

24. User-Controlled AI

Settings should include:

- Analyze all chats
- Analyze groups
- Analyze links
- Detect sensitive data
- Social engineering detection
- Attachment analysis
- Notification warnings
- Privacy mode

25. Chat Exclusion

Users can exclude chats from AI monitoring. The UI must clearly explain that excluded conversations will not receive normal SecureChat security analysis.

26. Temporary Privacy Mode

Allow:

- Pause AI for 15 minutes
- Pause until manually enabled

Useful for private conversations and testing.

27. Local AI Architecture

Prefer on-device processing:

Messaging App → Local Context Store → Lightweight AI Model → URL Analyzer → DLP Engine → Security Decision Engine

Only required/authorized processing should leave the device.

28. AI Memory Architecture

Use layered memory:

- Short-Term Context: recent messages
- Conversation Security Memory: relevant security observations
- User Security Preferences: trusted domains, blocked domains, sensitivity settings, excluded chats
- Long-Term Memory: only explicitly allowed information

29. Privacy-Preserving AI Memory

Each conversation has an independent security context. Avoid redundant centralized copies of private message history unless explicitly required and properly protected.

30. WhatsApp Companion Tab — SecureBridge

SecureBridge optionally analyzes notifications generated by other messaging applications, particularly WhatsApp.

Architecture:

WhatsApp → Android Notification → Notification Listener → SecureBridge → AI Security Agent → Risk Analysis → SecureChat Security Alert

31. Important WhatsApp Limitation

A notification-listener-based companion can process information exposed through notifications if the user explicitly grants notification access. It does not mean SecureChat can directly access WhatsApp's encrypted database.

Potential capabilities depend on Android permissions, notification settings, WhatsApp configuration, and OS restrictions.

32. Deleted Message Intelligence

If a notification containing message content was captured before a sender deletes the message, SecureBridge may preserve the content already exposed to the device, subject to user settings and applicable platform/privacy rules.

It must not claim to recover content that was never captured.

33. WhatsApp Companion Dashboard

Second tab can show:

- WhatsApp
- Critical alerts
- Suspicious messages
- Normal messages
- Organized notification history

34. Cross-App Security Notifications

If SecureBridge detects something dangerous:

"SecureChat Security Alert: A WhatsApp message appears to contain a high-confidence phishing attempt."

Actions:

- Inspect
- Ignore
- Block/Report if supported
- Open WhatsApp
- Add to Security Archive

35. Push Notification System

Examples:

Critical: "SecureChat blocked a high-risk link."

Suspicious: "SecureChat detected possible social engineering."

Sensitive data: "You may be sharing confidential information."

36. Smart Notification Priority

- Critical → Immediate
- High → Immediate
- Medium → Grouped
- Low → Silent
- Safe → No notification

37. Battery Optimization

Avoid sending every message to a large AI model.

Message → Cheap local classifier → Obviously safe? → Finish

If uncertain → Deeper analysis

This cascaded AI inference improves battery efficiency.

38. AI Inference Levels

Level 0 — Rule Engine: obvious URLs, OTPs, API keys, known indicators

Level 1 — Tiny ML Model: suspicious language, phishing patterns, spam

Level 2 — Contextual Model: uncertain cases

Level 3 — Advanced AI: complex cases only

39. Performance Requirements

Target:

- Basic message analysis: <100 ms
- Local security indicator: <200 ms
- URL heuristic analysis: <500 ms
- Deep analysis: <2–3 sec
- UI interaction: <100 ms perceived response
- Security notification: near real-time

Actual targets should be validated on representative devices.

40. Offline Security

Offline:

- Sensitive data detection
- Basic phishing heuristics
- Manipulation detection
- Local blocklists
- Known malicious pattern detection
- Local AI inference

Online:

- Threat intelligence
- Domain reputation
- Advanced URL scanning
- Model updates
- Security intelligence

41. SecureChat Messaging Features

Normal modern messaging capabilities:

- One-to-one chats
- Group chats
- Text
- Emojis
- GIFs
- Stickers
- Voice messages
- Images
- Videos
- Documents
- Contacts
- Location sharing
- Replies
- Forwarding
- Reactions
- Message editing
- Message deletion
- Disappearing messages
- Pinned messages
- Starred/saved messages

- Mentions
- Read receipts
- Typing indicators
- Message search

42. Groups

Support:

- Administrators
- Roles
- Permissions
- Invite links
- Approval system
- Group description
- Pinned messages
- Member management
- Security dashboard
- AI group context

43. AI Group Security

Examples:

- Malicious APK warning
- Sensitive information solicitation warning
- Suspicious external links
- Social engineering alerts

44. Attachment Security

Analyze:

- File type mismatch
- Dangerous extensions
- Suspicious archives
- Executable content
- Macro-enabled documents
- Malware indicators
- Suspicious PDFs
- Malicious scripts

Pipeline:

Attachment → File Metadata → Hash → Local Analysis → Reputation → Sandbox (if required) → AI Classification

45. Contact Trust System

Use communication security state rather than declaring people trustworthy.

Example:

Contact: Verified identity, 2 devices, 0 previous security incidents, 0 suspicious links, 0 credential requests, Trust state: Normal.

46. Zero-Trust Contact States

UNKNOWN → OBSERVED → KNOWN → VERIFIED

Security analysis remains active at every stage.

47. Account Compromise Detection

Detect unusual requests and behavioral anomalies. Example:

"Unusual conversation pattern. This request differs significantly from recent communication patterns."

Do not claim compromise without sufficient evidence.

48. Impersonation Detection

Identify:

- Newly created accounts
- Suspicious profile changes
- Username changes
- Unusual requests
- Suspicious links
- Identity inconsistencies

49. AI Explainability

Every warning should answer:

- What happened?
- Why is it suspicious?
- What evidence was used?
- How confident is the system?
- What should I do?

50. Security Center

Dashboard:

- Overall security

- Threats blocked
- Suspicious messages
- Phishing links
- Data leak warnings
- Manipulation alerts
- Recent security events

51. Personal Security Report

Weekly report can summarize:

- Phishing attempts
- Suspicious links
- Sensitive-data warnings
- Confirmed security incidents

Avoid unnecessary exposure of private message contents.

52. Security Event Log

Each event:

- Event ID
- Timestamp
- Conversation
- Risk Level
- Detection Type
- Confidence
- Detection Source
- Action Taken

53. User Privacy Dashboard

Show:

- What AI analyzed
- What data remains on device
- What data leaves device
- AI status
- Excluded chats
- Permissions
- Notification access
- Encrypted storage
- Security logs

54. "Why Did You Flag This?"

Provide concise evidence-based explanations, e.g.:

"I flagged this because the message combines an urgent account warning with a login link to a domain that doesn't match the claimed service."

55. AI Assistant

Users can ask:

- Is this safe?
- Why did you flag this?
- What is this person asking me to do?
- Does this link look legitimate?
- Am I revealing anything sensitive?
- Summarize the security concerns in this chat.

56. SecureChat AI Modes

- Guardian Mode — Maximum protection
- Balanced Mode — Recommended
- Privacy Mode — Maximum local processing
- Performance Mode — Minimal AI processing
- Custom Mode — User controls individual protections

57. Anti-Fatigue Security

Avoid excessive warnings. Learn from false positives and user feedback without automatically weakening critical protections.

58. Security Decision Engine

Risk can combine:

URL Risk + Message Risk + Context Risk + Sender Anomaly + Attachment Risk + Social Engineering Risk + Sensitive Data Risk + Threat Intelligence

Then:

Risk + Confidence → Security Policy → User Action

59. False Positive Management

Allow "This is safe" feedback. Use it for calibration, but do not blindly whitelist a sender. Critical threats should continue to be checked.

60. Architecture

Client:

- SecureChat Mobile
- Messaging UI
- Local Encryption
- AI Agent
- Risk Engine
- DLP Engine
- URL Engine
- Context Engine
- Secure Local Storage
- SecureBridge

Backend:

- API Gateway
- Authentication
- User Service
- Messaging Service
- Group Service
- Notification Service
- Security Intelligence
- Threat Intelligence
- Device Management

Infrastructure:

- PostgreSQL
- Redis
- Object Storage
- Encrypted APIs

61. AI Architecture

AI Security Agent:

- NLP Engine
- URL Engine
- DLP Engine
- Context Engine
- Risk Engine
- Explainability
- Security Decision

62. Suggested Backend Technology

Mobile:

- Flutter or native Android/Kotlin
- Android-native components for SecureBridge where necessary

Backend:

- TypeScript
- NestJS or Fastify
- PostgreSQL
- Redis
- WebSockets
- Object storage

AI:

- ONNX Runtime / TensorFlow Lite / ExecuTorch depending on selected models
- Lightweight local models
- Optional server-side inference for permitted tasks

Infrastructure:

- Docker
- Kubernetes later
- CI/CD
- Observability
- Encrypted secrets management

63. Core Database Entities

- User
- Device
- IdentityKey
- Session
- Conversation
- ConversationMember
- Message
- MessageAttachment
- MessageReaction
- MessageReply
- MessageReadReceipt
- Group
- GroupMember
- Contact
- SecurityEvent

- ThreatIndicator
- URLAnalysis
- AIContext
- AIObservation
- SecurityPolicy
- UserPreference
- ExcludedConversation
- NotificationCapture
- WhatsAppMessage
- PushNotification
- Report
- AuditEvent

64. SecurityEvent

Fields:

- id
- conversationId
- messageId
- userId
- type
- severity
- confidence
- source
- explanation
- recommendation
- createdAt

65. AIContext

Fields:

- id
- conversationId
- userId
- contextVersion
- summary
- riskState
- securityState
- updatedAt

Actual private message history should not be redundantly copied into centralized AI context unless explicitly required and properly protected.

66. Functional Requirements — Authentication

FR-AUTH-001: The system shall allow users to create SecureChat accounts.

FR-AUTH-002: The system shall authenticate registered devices.

FR-AUTH-003: The system shall support secure session management.

FR-AUTH-004: The system shall allow users to revoke active devices.

67. Functional Requirements — Messaging

FR-MSG-001: The system shall support one-to-one messaging.

FR-MSG-002: The system shall support group messaging.

FR-MSG-003: The system shall support message reactions.

FR-MSG-004: The system shall support replies.

FR-MSG-005: The system shall support message editing where permitted.

FR-MSG-006: The system shall support message deletion.

FR-MSG-007: The system shall support multimedia messages.

68. Functional Requirements — AI

FR-AI-001: The system shall provide a personal security agent to each user.

FR-AI-002: The agent shall maintain independent security context for each conversation.

FR-AI-003: The agent shall analyze messages for security risks.

FR-AI-004: The agent shall analyze URLs.

FR-AI-005: The agent shall detect potential phishing.

FR-AI-006: The agent shall detect social-engineering patterns.

FR-AI-007: The agent shall detect potential sensitive-data exposure.

FR-AI-008: The agent shall generate confidence scores.

FR-AI-009: The system shall distinguish inference from verified security evidence.

69. Functional Requirements — Warnings

FR-WARN-001: Low-risk messages shall display a green indicator when analysis is available.

FR-WARN-002: Suspicious messages shall display an orange indicator.

FR-WARN-003: High-confidence malicious messages shall display a red indicator.

FR-WARN-004: Critical events shall trigger configurable notifications.

FR-WARN-005: Users shall be able to inspect the reason behind warnings.

70. Functional Requirements — Zero Trust

FR-ZT-001: New communication sources shall be treated as untrusted by default.

FR-ZT-002: The system shall continuously evaluate security context.

FR-ZT-003: Trusted status shall not bypass security analysis.

FR-ZT-004: Users shall be able to manage trust-related settings.

71. Functional Requirements — DLP

FR-DLP-001: The system shall detect likely passwords.

FR-DLP-002: The system shall detect authentication codes.

FR-DLP-003: The system shall detect API credentials.

FR-DLP-004: The system shall warn users before sending potentially sensitive information.

72. Functional Requirements — WhatsApp Bridge

FR-WAB-001: The system shall provide an optional WhatsApp notification-analysis module on supported Android devices.

FR-WAB-002: The system shall require explicit notification-access permission.

FR-WAB-003: The system shall analyze captured notification content according to user configuration.

FR-WAB-004: The system shall preserve captured information only according to configured retention policies.

FR-WAB-005: The system shall never claim to recover notification content that was never captured.

73. Non-Functional Requirements — Security

Implement:

- Secure authentication
- Encryption
- Key management
- Least privilege
- Secure storage
- Secure APIs
- Rate limiting
- Abuse prevention
- Audit logging
- Secure update mechanisms

74. Privacy Requirements

- Data minimization
- Purpose limitation
- Local-first processing
- User transparency
- User control

75. Reliability

Target 99.9%+ production service availability with graceful degradation. If AI is unavailable, messaging continues and the security engine falls back to local protection.

76. Scalability

Architecture should support growth from 1,000 users to 100,000, 1 million, and 10 million+ users without a complete rewrite.

77. Threat Model

External:

- Account takeover
- Credential theft
- API attacks
- DDoS
- Malware
- Phishing
- Spam
- Malicious files

Internal:

- Unauthorized administrator access
- Insider abuse
- Database compromise
- Key compromise
- Logging leakage

AI-specific:

- Prompt injection
- Adversarial messages
- Context poisoning
- Model manipulation

- Malicious URLs designed to fool classifiers
- Adversarial attachments
- AI hallucinations
- False positives
- False negatives

78. AI Security Principle

Treat AI as an untrusted component.

AI Analysis + Deterministic Security Checks + Threat Intelligence + Policy Engine → Final Decision

79. AI Prompt Injection Defense

Conversation content must never be allowed to alter security-agent system policy.

User Content ≠ System Instructions

80. Security Explainability

Store evidence categories rather than exposing sensitive internal model reasoning.

Example:

Detection: Phishing

Evidence:

- Suspicious domain
- Credential request
- Urgency
- External login page

Confidence: 94%

Recommendation: Do not interact with the link.

81. Abuse Prevention

Controls:

- Spam detection
- Automated account creation detection
- Rate limiting
- Message throttling
- Abuse reporting
- Malicious attachment controls
- Bot detection
- Account reputation

82. Admin Security

Admin functions:

- System Health
- Security Metrics
- Threat Intelligence
- Abuse Reports
- Model Management
- Incident Management

Admin access must not automatically imply access to private chats.

83. Security Auditability

Every security-sensitive administrative action should generate an audit event with actor, timestamp, action, reason, and audit record.

84. Backup & Recovery

Support:

- Encrypted backups
- Database backups
- Key recovery mechanisms
- Disaster recovery
- Point-in-time recovery
- Backup integrity verification

85. Disaster Recovery

Define:

- RPO — acceptable data loss window
- RTO — acceptable recovery window

Values should be finalized during production planning.

86. UX Principle

The user should not feel like they are using a cybersecurity product.

Security should be:

- Invisible when safe
- Helpful when suspicious
- Impossible to ignore when critical

87. Main Navigation

Suggested:

- Chats
- SecureBridge
- Calls
- Security
- AI Guardian
- Settings

88. Chat Interface

Show:

- Contact name
- Security state
- Message content
- Security indicator
- Composer

89. Message Security Popup

Show:

SECURITY ANALYSIS

Risk Level

Confidence

Why?

Evidence

Recommendation

Actions: Inspect / Continue / Report

90. Security Center UI

Show:

- Overall Status
- Threats blocked
- Suspicious messages
- Phishing links
- Data leak warnings
- Manipulation alerts
- Protection toggles

91. Product Differentiators

1. Personal Security AI
2. Conversation-Level Intelligence
3. Zero-Trust Messaging
4. Social Engineering Detection
5. Data Leak Protection
6. Explainable Security
7. Cross-App Security
8. Local-First AI

92. Advanced Feature — Security Copilot

Guardian can answer:

- Is this person asking for something suspicious?
- What information have I revealed?
- Summarize risks in this conversation.
- Which messages should I be careful about?
- Is this link legitimate?

93. Advanced Feature — Secret Exposure Map

Classify shared information:

- Normal
- Personal
- Sensitive

Track credentials, financial data, identity information, etc.

94. Advanced Feature — Conversation Risk Timeline

Example:

- 8:01 PM Green — Normal
- 8:05 PM Green — Normal
- 8:09 PM Orange — External link
- 8:10 PM Orange — Urgency
- 8:11 PM Red — Credential request
- 8:12 PM Red — Phishing confirmed

95. Advanced Feature — Scam Pattern Library

Potential categories:

- Banking scams
- Courier scams
- Job scams
- Investment scams
- Romance scams
- University scams
- Government impersonation
- Account recovery scams
- Fake giveaways
- Crypto scams
- Tech-support scams

Use these as risk patterns, not as labels applied blindly to people.

96. Advanced Feature — Second Opinion

For uncertain messages:

"AI isn't certain."

Button: Get Security Second Opinion

97. Advanced Feature — Security Learning Mode

Explain why content was phishing or suspicious and teach users the attack pattern.

98. Development Phases

Phase 0 — Architecture:

- Requirements
- Threat model
- UX
- Cryptographic design
- Database design
- AI architecture

Phase 1 — Secure Messaging MVP:

- Registration

- Authentication
- Profiles
- Contacts
- 1:1 messaging
- Encryption
- Delivery
- Basic groups
- Reactions
- Replies
- Media

Phase 2 — Security Engine:

- Local classifier
- URL detection
- Phishing detection
- DLP
- Indicators
- Security events

Phase 3 — Personal AI:

- Conversation context
- AI summaries
- Contextual security
- Social engineering detection
- Explainability
- Security Copilot

Phase 4 — Advanced Protection:

- Attachment analysis
- Threat intelligence
- Account anomaly detection
- Scam detection
- Security center
- Reports

Phase 5 — SecureBridge:

- Notification listener
- WhatsApp notification processing
- Message organization
- Security analysis
- Notification archive

- Deleted-notification history where technically available

Phase 6 — Optimization:

- Model quantization
- Battery optimization
- Caching
- Inference routing
- Offline AI
- Performance optimization

99. MVP Definition

SecureChat MVP:

- Registration
- Login
- Profiles
- 1:1 messaging
- End-to-end encryption
- Basic groups
- Reactions
- Replies
- Media
- Local AI security engine
- URL phishing detection
- DLP
- Green/orange/red indicators
- Security explanations
- Push security alerts
- Security Center

V2:

- Personal conversation AI
- Social engineering detection
- Advanced threat intelligence
- Attachment analysis
- Account anomaly detection
- SecureBridge

V3:

- Calls
- Video

- Multi-device
- Desktop
- Advanced AI
- Enterprise security
- Advanced threat intelligence

100. Acceptance Criteria

Messaging:

- Reliable delivery
- Correct encryption
- Groups
- Replies/reactions
- Media

Security:

- Known phishing URLs detected with high reliability
- Sensitive-data warnings
- Correct security indicators
- High-confidence warnings
- Inspectable explanations

AI:

- Context isolation
- Prompt-injection resistance
- Inference vs verification distinction
- Messaging continues during AI degradation

Privacy:

- User controls monitoring
- Excluded chats remain excluded
- Private content minimized outside device
- Logs avoid unnecessary plaintext

Performance:

- Responsive messaging
- AI does not interfere with typing/scrolling
- Battery consumption is monitored and bounded

101. Most Important Architectural Decision

There are two products inside the concept:

Product A — SecureChat: a complete end-to-end encrypted messaging platform.

Product B — SecureBridge: an Android security companion that analyzes notifications from WhatsApp and potentially other messaging applications.

Keep them architecturally separated:

SecureChat Core ↔ Own Messaging / E2EE / Personal AI

SecureBridge ↔ Other Apps / Notifications / Security AI

This separation improves security, maintainability, and future expansion.

102. Final Product Architecture

SECURECHAT

- Messaging
- Guardian AI
- SecureBridge
- Zero Trust
- Phishing
- Social Engineering
- DLP
- Risk Engine
- Green / Orange / Red classification
- User Control
- Security Dashboard

Final Product Definition

SecureChat is a privacy-first, end-to-end encrypted messaging platform where every conversation is protected by a personal AI security agent that continuously evaluates messages, links, attachments, social-engineering patterns, and accidental data exposure under a Zero-Trust model—while SecureBridge optionally extends this protection to notification content from other messaging applications.

This is more than "WhatsApp + AI": it is an AI-powered personal communication security layer.